SYSTEM ARCHITECTURE OVERVIEW

Hyperlocal Multi-Vendor Fashion Marketplace

> Instant local fashion commerce connecting customers, sellers, and riders with a secure escrow-based settlement system.

TARGET	MODEL	SETTLEMENT
Men's Fashion	**Hyperlocal Delivery**	**T+7 Escrow**

CORE VALUE PROPOSITIONS

Hyperlocal Engine
Geo-search & nearby inventory fetching

Fast Delivery
Queue-based auto-assignment for riders

Trust & Escrow
Secure wallet with 7-day hold period

Strict QC Control
Multi-stage approval for Brands & Products

SYSTEM ARCHITECTURE

User Roles & Access Points

Four distinct user roles interact with the platform through dedicated interfaces, ensuring secure and efficient marketplace operations.

APP / WEBSITE

Customer

- Registration & Auto-Location Detection
- Browse Categories, Variants & Cart (Snapshot)
- Secure Payment & Real-time Tracking
- 7-Day Return Window Management

SELLER PANEL

Seller

- KYC Submission & Store Verification
- Brand, Product & Variant Creation (QC Flow)
- Order Management & Stock Control
- Wallet Payouts (T+7 Settlement)

RIDER APP

Rider

- Onboarding, KYC & Online Status Toggle
- Auto-Assigned Orders (Geo + Queue)
- Pickup & Delivery Fulfillment

CONTROL PANEL

Super Admin

Approvals: Sellers, Riders, Brands & Products

Category, Attributes & Commission Mgmt

Order Monitoring, Settlements & Refunds

PROCESS FLOW

End-to-End Order Lifecycle

Customer | System | Seller | Rider | Wallet

1 Discovery & Order

Geo-Location
Fetch nearby sellers

Add to Cart
Price snapshot stored

Payment
Order Placed

2 Orchestration

Order Processing
Stock Reserved

Broadcasting
Notify Seller & Search Riders

3 Assignment & Prep

Seller Prep
Accept & Pack Order

Auto-Assign
Queue-based engine

4 Delivery

Pickup
Navigate to Store

Delivered
Customer Handover

5 Settlement

7-Day Hold
Return Window

Escrow Release

FINANCIAL ARCHITECTURE

Payment & Wallet Escrow Flow

Escrow Secured | T+7 Settlement

Customer Pays ₹1000
Payment Gateway Success

SELLER	PLATFORM	RIDER
₹850	₹50	₹100
LOCKED	LOCKED	LOCKED

7-DAY RETURN WINDOW

FUNDS RELEASED
Auto-settled to wallets

SETTLEMENT LIFECYCLE

STAGE	PLATFORM	SELLER	RIDER	STATUS
Payment Received	+₹1000	0	0	HOLDING
Order Delivered	Locked	Locked	Locked	ESCROW
After 7 Days	-₹900	+₹850	+₹100	SETTLED

REFUND & EXCEPTION HANDLING

Cancel Before Pickup
Customer cancels immediately
Full Refund to Customer
Optional gateway charges

Cancel After Pickup
Item already with rider
Refund minus charges
Small fee credit to Rider

Not Delivered
Customer unavailable
Full Refund to Customer
No pay for Seller/Rider

Shop Didn't Ship
Inventory error / Seller fault
Full Refund to Customer
Deduct charges from Seller

Return After Delivery
Within 7-day window
Full Refund to Customer
Reverse logistics initiated

QUALITY ASSURANCE

QC Workflow & Approval Timeline

Pending Approved Rejected

Product Level QC

ID: PROD-FLOW-01

Seller
Create / Update Product

→

Pending
Not Visible to Customer

→

Super Admin
Review Details & Images

Live / Visible
If variants approved

Rejected
Reason note sent

Variant Level QC

ID: VAR-FLOW-02

Seller
Add Size / Color Variant

→

Pending
Inventory Locked

→

Super Admin
Check Specs & Price

For Sale
Stock available

Rejected
Reason note sent

LOGIC GATES

Visibility Rule

Product creates parent shell. It is HIDDEN until at least one Variant is APPROVED.

Rejection Feedback

Admin selects reason code + types note. Seller sees specific reason in panel to fix & resubmit.

Audit Trail

All status changes (Pending → Live → Rejected) are logged with Admin ID and timestamp.

TARGET METRICS

Avg Approval Time	< 24h
Auto-Reject Rate	0%

ARCHITECTURE & ROADMAP

System Modules & Future Scalability

Core Active

Planned (Q3)

USER & ROLES

- Customer Profile
- Seller Panel
- Rider App
- Admin Controls

CATALOG

- Brand
- Wear-Type
- Cat / Sub-Category
- Product (QC)
- Variant (QC)

COMMERCE

- Cart (Snapshot)
- Order Management
- Return / Refund
- Stock Reservation

FINANCE

- Wallet System
- Transaction Ledger
- Settlement Logs
- Escrow Logic

FUTURE SCALABILITY ROADMAP

Microservices
Decoupled Architecture

Redis Sessions
Single Sign-On Speed

Queue Workers
Kafka / BullMQ

Real-Time Ops
Live Map Tracking

RISK FLAGS

- High Return Customer
- Fake/Ghost Seller
- Multiple Cancellations
- Suspicious Rider Data